

Sleep State Classification Using EEG Data

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Final project Review Paper

Sleep is a critical part of human health, but measuring sleep efficiency and behavior can be a complicated process. Polysomnography (PSG) is the gold standard in sleep monitoring but there are several issues with it that prevent it from being more useful. The process is time consuming and invasive (many sensors are needed to pick up the electrical signals from the body). The analysis to classify the sleep stages of a patient is done manually and in retrospect, making it impractical. Also, experts often disagree on how to label the data. Automated systems have been developed but they are seldom used in clinical settings and are more applicable for adults. One of the settings where automated sleep scoring would be very valuable is for the pediatric intensive care unit (PICU), but current models have not been validated for this use. One of the obstacles is that critically ill children face complications that affect the data like developmental changes in sleep patterns. Even when not used in clinical settings the polysomnography can pose issues. Besides the disadvantages of manual classification of data, many of the issues come from using multi-channel systems to capture data, which adds unneeded complexity and limits the movement of animal subjects. Free movement of the subjects is vital data because sleep stages can be classified based on the degree of physical movement.

Scientists have thus been working on a system for classifying sleep stages that would be more practical in any setting (clinical or for animal and neuroscience studies). One of those systems involves using EEG signals. An EEG is an electrical signal from brain waves that is non-invasive in humans and has the potential for a model to interpret the data in real time. Smith et al. (2021) had scientists working with rodents to develop an automated classifier that accurately predicts sleep states using only single channel EEG data. Several different classification models are tested such as decision trees and random forests. The researchers hypothesize that high accuracy in sleep state classification can be achieved with relatively low input data (using data from the EEG).

Van Twist et al. (2024) focuses on pediatric sleep state classification. The researchers also work with single channel EEG data, but they work to develop a sleep index to monitor the sleep of pediatric patients. They hypothesize that by using gamma to delta power ratios to characterize sleep depth combined with a machine learning model will produce accurate classification and generalizability to new data.

Both studies work on the challenge of interpreting less data, and to replace PSG scoring with automated machine learning. However, Smith et al. (2021) has the goal of increasing research efficiency with improved accuracy while Van Twist et al. (2024) aims to increase clinical deployment of sleep scoring through greater interpretability.

Smith et al. (2021) worked with EEG, EMG, and movement data, totaling 427 hours of activity data. EEG was used to capture brain signals and EMG was used to capture muscle activity signals. The data was split into small time chunks of 10 seconds (1 epoch) for the model to analyze piece by piece, with experts manually labeling each as either REM, slow-wave (deep sleep), or wake. From each segment the data was converted into numerical features for the model to classify the sleep stages. There are 42 features for each 10 second segment, 40 for the EEG

data, 1 for EMG data, and 1 for movement data. As a result, there was extensive data with a great amount of detail. The researchers used 5 consecutive segments, totaling 210 features. Having several consecutive segments reduces the noise of the data and gives temporal context that reduces misclassification. The manual expert labels were compared to the automated labeling done by the model to measure accuracy.

The researchers classified the data by comparing several models that included Decision Trees, Random Forests, Naïve Bayes Classifiers, Logistic Regression Classifiers, and Artificial Neural Networks(ANN). These models were chosen in order to determine which ones are the best classifiers for the job based on their accuracy in correctly labeling the data. The most accurate models were the ANN and random forest. Decision trees are pivotal to the random forest model. Decision trees classify data through sequential decisions of feature thresholds, like using cutoffs to branch off samples based on their values. A random forest is multiple decision trees working together, where each tree makes its own prediction, and the majority decides the output. Random forests are thus more accurate than a decision tree but harder to interpret. On the other hand, ANNs work with layers of artificial neurons that process information through weighted values. Random forests and ANNs are good at recognizing complex, non-linear relationships in the data, making them the most effective models for animal and neuroscience studies.

Van Twist et al. (2024) uses EEG data from 90 children, converting the raw signal into frequency based features that measure the brain activity at different speeds (fast and slow brain waves). These are known as spectral band powers and are used to distinguish between sleep stages. The researchers used 30 second segments (epoch) and basically only used 1 feature, the gamma delta ratio based on the EEG data. Gamma waves are faster and indicate an active brain (wake), whereas delta waves are slow and indicate deep sleep. The ratio tells us the likely brain state the subject is in. The researchers smoothed out their data by averaging nearby segments, and similarly to using 5 consecutive segments in the other paper, it reduced noise and improved accuracy. After converting raw EEG data into a number value called a sleep index, a decision tree is used to answer cutoff questions from the sleep index value. Decision trees were chosen because they are relatively simple and easy to interpret, making it a viable method for classifying sleep stages in a clinical setting.

Smith et al. (2021) had high performance especially for ANN and random forest, falling at around 85-95 % accuracy. The major way performance of the model was evaluated was with accuracy, measuring the percent of predictions made by the model that were correct compared to human annotations.

Smith et al. (2021) had many strengths in their model used, one of which was including multiple forms of input data such as EEG, EMG and movement data, capturing more useful data to classify sleep. Another strength is that the dataset was large, including 427 hours of sleep data. In addition, ANN and random forest models are more advanced than other models, allowing for better capturing of complex patterns. Finally, using an automated system saved time and prevented human error. However, one of the many limitations of their model was that by using multiple sensors and advanced classification models the data was harder to interpret and it was more computationally intensive. The research was also done on rodents, lowering the degree of human application.

For Smith et al. (2021), the deployment of their research is limited to research labs and animal

sleep studies due to the high degree of equipment complexity and high computational demands. While limited, the research done has advanced the field because it is based on improved automated sleep scoring methods and demonstrates how to handle large biological datasets with advanced machine learning classification models.

Van Twist et al. (2024) uses balanced accuracy to evaluate two, three, and four sleep states, resulting in accuracies of 0.88, 0.74, and 0.57 respectively. Therefore, the model works well for simple classifications and less so for more detailed sleep stages. Their model has many strengths. By using a single EEG channel and decision tree with a sleep index, the data is more simple and the model is easily interpretable. As a result, their model is applicable to clinical settings. There are also several limitations to the model. Complex classification showed low accuracy (4-state sleep). Using gamma delta ratio to simplify the data into an index loses detail. The model was also only used by non-critically ill children which reduces generalizability. The model is most applicable in the PICU, as a potential real time tool for sleep monitoring of patients. Their research advances the field because it aids in the development of a fast, highly accessible, and non-invasive model to monitor sleep in hospitals.

Both studies are trying to improve automated sleep classification using machine learning. Smith et al. (2021) is relevant to neuroscience research, contributing to the goal of analyzing large animal datasets faster while using a high dimensional approach. Their research develops more accurate models for automated sleep scoring, exploring single channel EEG and EMG, as well as activity data, random forest and ANN models. Van Twist et al. (2024) is focusing on developing a model for clinical usability and interpretability based on simple 2-3 sleep state classification, opting for a low dimensional approach with a simple model (decision tree).

References

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